Calibration of Tellus 2D, an agent-driven plane-strain geotechnical finite element program

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Abstract

Tellus 2D is a plane-strain geotechnical finite element program whose primary user is a large language model (LLM) agent, which builds, calculates and reports a model through 33 typed tools of the Model Context Protocol (MCP). This paper calibrates the program, with every model run headless through its MCP server, against 82 verification tests, three lessons of the PLAXIS 2D Tutorial Manual and four instrumented case histories in soft clay, using the published parameters without tuning. The verification results agree with closed-form and reference solutions within 4%, and the tutorial factors of safety agree with PLAXIS within 4.3%. In the field the program reproduces the wall deflection of a 21 m Bangkok excavation within 13% and the final settlements of a vacuum-preloading trial within 3%, but fails the Muar embankment 1.2–1.4 m early, as its three-dimensional sibling Tellus3D does. The embankments on soil-cement columns fall within the measured scatter only with the column modulus of the source, so the remaining differences come from uncertain inputs rather than from the code. The calibration found and fixed four defects that the existing tests had missed, and all 30 analyses ran in 1.8 hours on one RTX 4090 with the GPU solver ported from Tellus3D.

Keywords finite element method · geotechnical engineering · verification and validation · PLAXIS tutorial · AI agent · Model Context Protocol · consolidation · deep excavation

1 Introduction

Geotechnical finite element (FE) programs are trusted because they have been compared, over many years, with closed-form solutions, with each other and with field measurements, and a new program has to earn the same trust before its numbers can enter design. Tellus 2D is such a new program, a plane-strain code written in Rust with the usual soil models and structural elements, and its distinguishing feature is that it is designed for an AI agent as its primary user [5, 6]. An agent drives the program through typed tools, building the soil profile, the structures and the construction phases, starting the calculation, reading results at points and writing a PDF report, while the engineer reviews the model and the report rather than drawing the model. This division of work makes the calibration of the solver more important rather than less, because an engineer who did not build the model has fewer chances to notice a number that is wrong.

The calibration of a program is usually reported as a verification against problems with known answers and a validation against measured behaviour [7]. Verification shows that the equations are solved correctly, whereas validation shows that the equations and their parameters represent the ground. Between the two lies a comparison with an established program on fully specified problems, which tests the same equations on the same inputs in the combinations that practice uses, staged excavation, consolidation and strength reduction, without the uncertainty of the field. Each step answers a different question, so a calibration needs all three, and it needs them across the applications that the program will

be used for, because a code that is accurate for a footing can still fail in a staged consolidation or an excavation with interfaces.

This paper reports all three steps for Tellus 2D in eight applications, from bearing capacity, slope stability, consolidation, groundwater flow and piles to deep excavation, vacuum preloading with vertical drains and embankments on soil-cement columns. The verification uses 82 automated tests against closed-form solutions, a reference FE code and published element tests, the comparison with an established program repeats three lessons of the PLAXIS 2D Tutorial Manual [1] with the manual's inputs, and the validation repeats four instrumented case histories in soft clay, three of them used earlier for the three-dimensional sibling program Tellus3D [8] and one of embankments on soil-cement columns, all with the published parameters and no tuning to the field data. Every model was built and run headless through the program's MCP server by an LLM agent, as the program is meant to be used, and the final runs of all 30 analyses were made one after another on a cloud GPU, so the paper also reports what such a calibration costs. The calibration found four defects that the existing tests had missed, and it changed the program, so the changes and the reasons for them are reported together with the comparisons.

Section 2 summarises all benchmarks and their results in one figure and one table, and the sections after it give the details. Section 3 describes the program and the way the cases were run, Section 4 lists the changes that the calibration made necessary, Section 5 gives the verification suite, Section 6 repeats the three tutorial lessons, Section 7 repeats the three case histories of Tellus3D, Section 8 adds the case history of soil-cement columns, Section 9 reports the computing times, and Section 10 discusses what the comparisons show and where their limits lie.

2 Summary of the benchmarks

Tellus 2D reproduces the closed-form and reference solutions within 4% in every one of the 14 verification results of Figure 1a, and 9 of the 13 results of the PLAXIS tutorial lessons within 10%, whereas against field measurements its accuracy depends on the application. Table 1 lists the eight applications, the calculations and models that each uses and the result of its benchmarks, and the sections that follow give the details. The factors of safety of the stability problems agree with the references within 4.3%, the consolidation and flow solutions within 3%, and the vacuum trial reproduces the measured final settlements within 3% at three depths. The largest differences are in the field cases whose inputs the sources do not fix: the Muar embankment fails at 74–79% of its measured failure height, and the settlements of the soil-cement columns lie at 0.65–1.64 times the measured group means with the column modulus of the source but at 0.25–0.56 times with the rule $E = 100\ q_u$. These differences come from the inputs and the constitutive models rather than from the program, because Tellus3D makes the same errors on the same data wherever both programs were run.

All 30 analyses of the paper ran on one RTX 4090 in 1.8 hours, from 11 s for a tutorial excavation of 706 elements to 18 minutes for a soil-cement column model of 8,650 elements and 207,535 equations (Figure 1b). The time grows with the size of the system and with the number of consolidation steps, since each change of the time step needs a new factorisation, and it is the column models, whose thin walls refine the mesh, that take longest. The calibration also found four defects in the program, which Section 4 describes, and three of them, in the interfaces, the start of safety analyses and the relaxation of consolidation phases, had made benchmarks of this paper fail before they were fixed.

3 The program and how the cases were run

Tellus 2D solves plane-strain equilibrium with 15-node cubic-strain triangles, which are accurate in collapse and in nearly incompressible undrained loading [9], and optionally with 25-node rectangles on a structured grid that conform with the triangles. The soil models are linear elastic, Mohr–Coulomb with a tension cut-off and the undrained types A, B and C, Hardening Soil [10] with a cap calibrated to K_0^{nc} and E_{oed}^{ref}, HS small [11], Soft Soil and Soft Soil Creep [12] and the hypoplastic sand model of von Wolffersdorff [13]. Structures are Mindlin plates with zero-thickness interfaces on either side, spring supports, geogrids and embedded piles. The groundwater is hydrostatic below a phreatic line, follows a measured profile or comes from a steady-state flow analysis, consolidation is fully coupled [14], and a φ–c reduction gives the factor of safety. The phases are solved by load stepping with the initial-stiffness method, Aitken relaxation and an automatic step size, and, as in PLAXIS, a step is accepted when the out-of-balance is below 1% of the reference load, or, after all iterations, when it is below 3% and its excess over 1% sits at a few nodes.

The program is a desktop application and, with the flag `--mcp`, a headless MCP server on standard input and output.

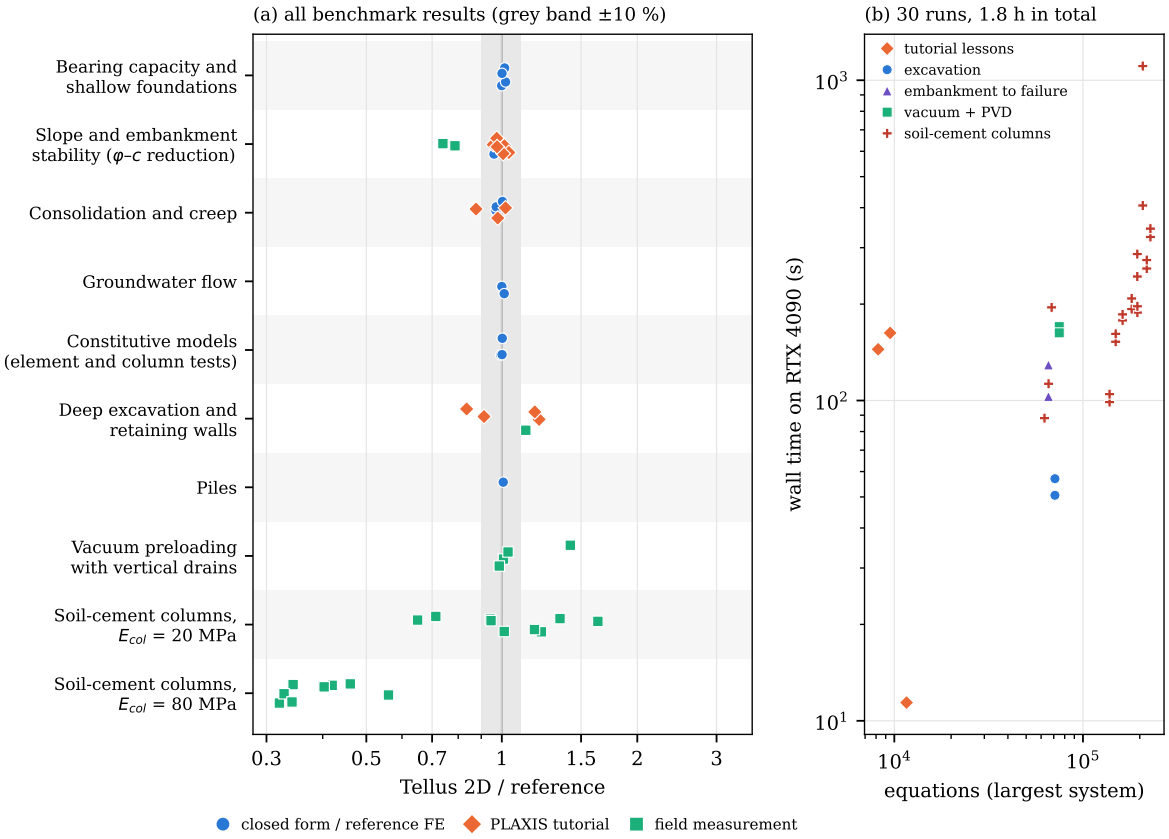


Figure 1: Summary of the calibration. (a) Every benchmark result of the paper as the ratio of Tellus 2D to its reference, grouped by application and coloured by the kind of reference: closed-form solutions and a reference FE code (verification, Section 5), the PLAXIS 2D Tutorial Manual (Section 6) and field measurements (Sections 7 and 8). The grey band is ±10%, the points of a row are spread vertically to show them all, and the soil-cement column results are ratios to the mean of each measured group. (b) Wall time of each of the 30 runs on the cloud box against the largest system of equations it factorised (Section 9).

Each case of this paper is a Python script that starts the server, builds the model through the same tools an agent uses, calculates, reads the results at the instrument positions with `get_results` and `get_curve`, and writes the deformed meshes as images with `show_picture`, so that every figure of a mesh in this paper was drawn by the program during its run. Running headless keeps the memory to that of the model and the solver, so several cases ran side by side. The cases were developed on a laptop (Intel Core i9-9880H, eight cores, 32 GB), and the final runs of all of them, whose results this paper reports, were made one after another on a rented cloud box (AMD EPYC 7502 and an NVIDIA RTX 4090) reached with the route of Section 9. The scripts record every parameter and its source, and the field cases use the parameters, geometry and assumptions of the Tellus3D benchmarks, so the two programs see the same data.

4 Changes made during the calibration

The calibration changed the program, and the changes are reported because they are its most useful result. Four are defects that the 75 tests of the existing suite had not found, five are features or consistency changes that the cases needed, and the last brought the solver of Tellus3D to the program.

Defect: interfaces on an excavated side. An interface duplicates the plate's nodes on its side and joins them to the plate through springs that carry an effective stress. It acted whenever its plate was active, so when the soil in front of a wall was excavated the interface kept the K_0 normal stress it had been given and went on pushing the wall back into the retained soil. In the excavation of Section 7.1 this put a roof slab in 26 kN/m of tension where it should have been in

Table 1: Applications calibrated in this paper, the calculations and models they use, their benchmarks and the agreement. FoS: factor of safety; HS: Hardening Soil; SS: Soft Soil; MC: Mohr–Coulomb.

Application	Calculation and models	Benchmarks	Agreement
Bearing capacity	plastic, MC and HS, undrained B and C, φ–c reduction	footing (reference FE), strip collapse $(2+\pi)s_u$, strip FoS	within 2%
Slope and embankment stability	φ–c reduction, drained and undrained A, HS and SS	Griffiths–Lane slope, PLAXIS Lessons 2 and 9, Muar to failure	FoS within 4.3%; Muar fails 1.2–1.4 m early
Consolidation and creep	Biot consolidation, SS and SS Creep	Terzaghi, creep settlement, PLAXIS Lesson 9	within 3%; Lesson 9 peaks −2 and −12%, dissipation 3 times faster
Groundwater flow	steady confined and unconfined flow	discharge of an aquifer and a dam	within 1.2%
Excavation and retaining walls	staged excavation, HS, plates, struts, interfaces, pore pressure profiles	PLAXIS Lesson 3, Sukhumvit 21 m excavation	9–17% dry, 18–21% submerged; Sukhumvit +13%
Piles	embedded pile with skin and base capacity	axial capacity	within 0.7%
Vacuum preloading	consolidation with vacuum, PVD equivalent permeability, SS	Suvarnabhumi TV2	within 3% at three depths
Soil-cement columns	column walls, undrained lifts and consolidation, SS and MC	Lian-Yun-Gang, nine measured groups	0.65–1.64 of group means (E_{col} = 20 MPa)

compression and collapsed the second stage. An interface segment now acts only while all of its soil-side nodes belong to active soil, and a test excavates 3 m in front of a cantilever sheet pile with and without interfaces, which now moves the wall top 3.07 and 3.21 mm towards the pit.

Defect: lines along cluster edges. A line load or plate that lay along part of a cluster edge was split into points twice, at two spacings, which left nearly coincident points on one constraint line and made the refinement create slivers without limit. A river-bank model with a 5 kPa mattress load along part of its dredged slope meshed to 185,000 elements instead of about 4,600. A piece of line that lies on a finer line is now left to that line, and a test checks that a mattress load along a dredged slope leaves the mesh at its size (4,442 against 4,481 elements).

Defect: the start of a safety analysis. A safety phase applied the out-of-balance that the phase before had left within its tolerance together with the first strength reduction. When that leftover sat at a few nodes that were already at failure, as at the toe of the embankment of Lesson 9 after consolidation, no strength step could resolve it and the analysis stopped at $\Sigma M_{sf} = 1.000$ without a step, and when it acted as an unloading it raised the factor of safety, to 3.7 for a footing on Hardening Soil sand whose Mohr–Coulomb counterpart has 1.46. A safety phase now first resolves the leftover at full strength and then reduces the strength from that equilibrium, and it leaves a leftover that cannot be resolved as it was. This gives the factors of safety of Section 6, the Hardening Soil footing 1.50, and leaves the Prandtl strip and the Griffiths and Lane slope unchanged (2.041 and 1.322, Table 2), but it moves one slope without a reference (Section 10).

Defect: unrelaxed consolidation phases. The iterations of a plastic phase relax each correction with the Aitken factor, which damps stress points that oscillate between yielding and unloading, but those of a consolidation phase did not. In the soil-cement column case (Section 8) a few such points beside the top edge of a column wall or in the soft clay under a fill held the error at about four times the tolerance, so three long consolidation phases stopped as collapses at 0.2–12% of their time, although nothing was failing, and the analysis of TV2 with the calibrated vacuum stopped at day 9.2. With the relaxation in consolidation phases too, all four run to the end, while a consolidation that converged before is unchanged, as Lesson 9 of Section 6 shows with the same pore pressures and settlement and factors of safety that move by at most 0.02.

Features. A water level can carry a measured pore pressure profile $u(y)$, capped by hydrostatic pressure below its own line, $u = \min\{u(y), \gamma_w(h(x) - y)\}$, so a level lowered into a pit dewaters the pit while the regional drawdown

governs outside it. A consolidation phase can hold the porous nodes inside a box at a prescribed suction, ramped over the phase, for a vacuum membrane or a drain, and the test applies 100 kPa to the drained top of an elastic layer and recovers Terzaghi's degree of consolidation and the one-dimensional settlement. Open water now presses on a plate face that has no active soil in front of it, which the submerged excavation of Lesson 3 needs, because the water left in the pit pushes on the wall. Interfaces in undrained soil are now undrained, as in PLAXIS, with an excess pore pressure $\Delta p = (K_w/n)\,\Delta u_n/t$ from the interface's normal opening over its virtual thickness in plastic phases and the excess of the adjacent soil node in consolidation phases, so an undrained unloading reduces their effective normal stress and strength as little as it does in the soil. Finally, the Soft Soil model now integrates the stress in substeps of at most 5% of p' and both Soft Soil and Hardening Soil floor the iteration stiffness at a reference stress (30 kPa and 0.3 p_{ref}), as Tellus3D does, because near-surface clay under vacuum made the iterations oscillate without them.

Solver and cloud route. The scalar up-looking $\mathrm{LDL}^{\mathrm{T}}$ factorisation was replaced by the supernodal multifrontal Cholesky and $\mathrm{LDL}^{\mathrm{T}}$ of Tellus3D [3, 4], which factorises independent subtrees in parallel with dense GEMM updates, and by NVIDIA cuDSS [2] when a GPU is present, loaded at run time so that the same binary runs without it. The results became serialisable, a `tellus --http` server answers MCP over HTTP and calculates projects sent to it, and a script rents a cloud GPU box, builds the program there and sets it as the compute server of the app, the chat and the local MCP server. `show_picture` can also write a picture to a file, with a window, a displacement scale and a reference phase, and plates that are not active are no longer drawn.

5 Verification

The verification suite has 82 tests in the solver crate, of which 77 run by default and five long ones on request, and Table 2 lists a selection. All of them pass on the final version, in under 2 min on the cloud box, and of the 69 tests of the desktop application all but the one discussed in Section 10 pass. The suite covers the elastic response against a reference FE code, consolidation against Terzaghi's solution [15], undrained bearing capacity against Prandtl's solution, the slope of Griffiths and Lane [16], steady seepage, the Hardening Soil and HS small models against their defining equations and a reference code's element tests, Soft Soil Creep against its creep law, the structural elements, and the five tests added in this work. The suite also shows the known weaknesses, because one-way consolidation reaches $U = 0.490$ at $T_v = 0.2$ against 0.505, about 3% slow at that time, and the slope factor of safety is 4% below Bishop's value, on the safe side.

6 Lessons of the PLAXIS 2D Tutorial Manual

The PLAXIS 2D Tutorial Manual [1] gives fully specified problems with their results, so Tellus 2D can be compared with an established program on the same input, without the uncertainty of field data. Three of its plane-strain lessons were repeated with the manual's geometry, materials, phases and mesh density (Medium or Fine), and their results are compared in Table 3 and Figure 2. The manual gives most results as figures, so its values were read off them to about 2–3% of each axis. The other lessons need features that Tellus 2D does not have and were left out: Lesson 1 is axisymmetric, Lessons 7, 8 and 10 need prestressed node-to-node anchors or wells, and the later lessons need transient flow, rock joints, dynamics or thermal analysis.

Lesson 2: drained and undrained stability of an embankment. A 4 m embankment of Hardening Soil sand stands on 6 m of Hardening Soil clay with the water at the ground surface, and the factor of safety is found by φ–c reduction after construction on drained and on undrained (type A) clay. Tellus 2D gives 1.833 and 1.330 against 1.81 and 1.39, and the mechanisms are those of the manual, a slip through the embankment slope on drained clay and a deep failure of the clay that spreads the embankment on undrained clay (Figure 3a, b). Tellus cannot change a material between phases, so the drained and undrained cases are two models.

Lesson 3: submerged construction of an excavation. A 20 m excavation, 30 m wide, is retained by 30 m diaphragm walls with interfaces of $R_{\mathrm{inter}} = 0.5$ in clay and 0.67 in sand and one strut at 1 m depth, in 20 m of undrained Hardening Soil clay over drained sand, and the second and third excavation stages are carried out under water, which stays in the pit and presses on the wall and the base. The wall displacement at (50, 10) follows PLAXIS within 9–17% through the load, the first excavation and the strut (Figure 2a), but it is 21% and 18% larger after the two submerged stages, 68.4

Table 2: Selected verification tests of Tellus 2D on the final version. Reference FE values are those of a commercial 15-node code on the same problem.

Test	Tellus 2D	Reference
Footing on sand over clay, 50 kPa, max $\|u\|$	9.12 mm	9.125 mm (reference FE)
Terzaghi consolidation, U at $T_v = 0.2$ (one-way)	0.490	0.505 (analytical)
Terzaghi consolidation, U at $T_v = 0.8$ (two-way)	0.862	0.887 (analytical)
Vacuum of 100 kPa on the drained top, U at $T_v = 0.2$; final settlement	0.490; 1D value	0.505; $p_{vac}H/E_{oed}$
Strip on undrained clay, collapse load, Undrained B / C	104.8 / 104.3 kPa	$(2+\pi)s_u$ = 102.8 kPa
Strip at half its collapse load, factor of safety	2.041	2.039
Griffiths and Lane 2:1 slope, factor of safety	1.322	1.376 (Bishop), 1.4 (their FE)
Confined steady flow, discharge	1.00000 m^3/day/m	1 m^3/day/m
Unconfined flow through a rectangular dam, discharge	3.037 m^3/day/m	3.0 (Dupuit–Charny)
Pore pressure profile, also under a lowered line, at every stress point	to 10^{-6} kPa	the capped profile
Hardening Soil, 10 m column under 100 kPa	30.46 mm	30.39 mm, $\iint d\sigma/E_{oed}$
Hardening Soil, triaxial failure in compression and extension	−372.86 / −26.06 kPa	−372.86 / −26.06 kPa (reference FE)
Soft Soil, one-dimensional loading, K_0 and ε_v	0.5967, 0.0800	0.6, 0.0799
Soft Soil Creep, 10 m layer, settlement from 100 to 1,000 days	0.0920 m	0.0917 m, $\mu^* H \ln(t_2/t_1)$
Embedded pile at 110% of its capacity	fails at 110.8 kN/m	110 kN/m
Cantilever wall excavated 3 m, wall top with / without interfaces	3.07 / 3.21 mm	close, towards the pit
Line load along part of a slope edge, elements with / without it	4,442 / 4,481	no runaway refinement
Supernodal LDL^T of a quasi-definite system, residual	$< 10^{-8}$	0

and 168 mm against 56.5 and 142 mm. Three of the changes of Section 4 were needed to reach this result. Without the water pressure on the wall the second stage deflected 96.8 mm and the third collapsed, and with it but with drained interfaces the two stages gave 75.5 and 181 mm. The remaining difference is not the mesh (75.2 mm on a mesh three times finer) nor the interface stiffness (no change when the clay's elastic modulus is varied five-fold), and it disappears when the clay interfaces are given the soil's full strength ($R_{\text{inter}} = 1$ gives 56.6 and 151 mm). It therefore lies in the response of weak interfaces against undrained Hardening Soil clay, where the undrained Hardening Soil of Tellus is already known to depart from that of PLAXIS by up to 15% in element tests.

Lesson 9: construction of a road embankment. A 4 m embankment with 1:3 slopes is built in two lifts on 3 m of peat and 3 m of clay (Soft Soil, undrained A) over dense sand, with consolidation phases of 2, 30, 1 and 22.5 days and a safety analysis after each, the last of them ending at day 55.5 where the manual's consolidation to a minimum excess pore pressure of 1 kPa ends. The excess pore pressure under the centre peaks at 25.1 and 27.9 kPa after the two lifts against 25.8 and 31.7 kPa, and the settlement of that point reaches 0.217 m against 0.213 m, but the pore pressures dissipate about three times faster than in PLAXIS, with a half-life of about 2 days against 5 days after the first lift, so the settlement reaches its plateau earlier (Figure 2b, c). Part of the difference is that PLAXIS lowers the permeability with the void ratio (its $c_k = 1.0$ for peat and 0.2 for clay), which Tellus does not, and which accounts for a factor of about 1.5. The likely remainder is that PLAXIS reduces the permeability of the metre of peat above the water table with its unsaturated relative permeability, which impedes upward drainage, whereas Tellus treats all porous soil in a consolidation phase as saturated. The four factors of safety are 1.130 / 1.744 / 1.158 / 1.425 against 1.09, 1.79, 1.15 and 1.46, within 3.6%, and the mechanism after the second lift runs through the soft layers beneath the slope as in the manual (Figure 3f). These factors of safety became possible only with the corrected start of a safety analysis, because before it every one of them stopped at 1.000.

Table 3: PLAXIS 2D Tutorial Manual lessons in Tellus 2D. PLAXIS values read off the manual's figures (2–3% of each axis).

Lesson and result	Tellus 2D	PLAXIS	Difference
2: factor of safety, embankment on drained clay	1.833	1.81	+1.3%
2: factor of safety, embankment on undrained clay	1.330	1.39	−4.3%
3: wall $\|u\|$ at (50, 10) after the load / first excavation / strut	5.0 / 22.4 / 22.4 mm	6 / 24.5 / 24.5 mm	−17 / −9 / −9%
3: after the second (submerged) / third excavation	68.4 / 168 mm	56.5 / 142 mm	+21 / +18%
9: excess pore pressure at (0, −3), peaks of the two lifts	25.1 / 27.9 kPa	25.8 / 31.7 kPa	−3 / −12%
9: settlement at (0, −3) at day 55.5	0.217 m	0.213 m	+2%
9: factor of safety after phases 1 / 2 / 3 / 4	1.130 / 1.744 / 1.158 / 1.425	1.09 / 1.79 / 1.15 / 1.46	+4 / −3 / +1 / −2%

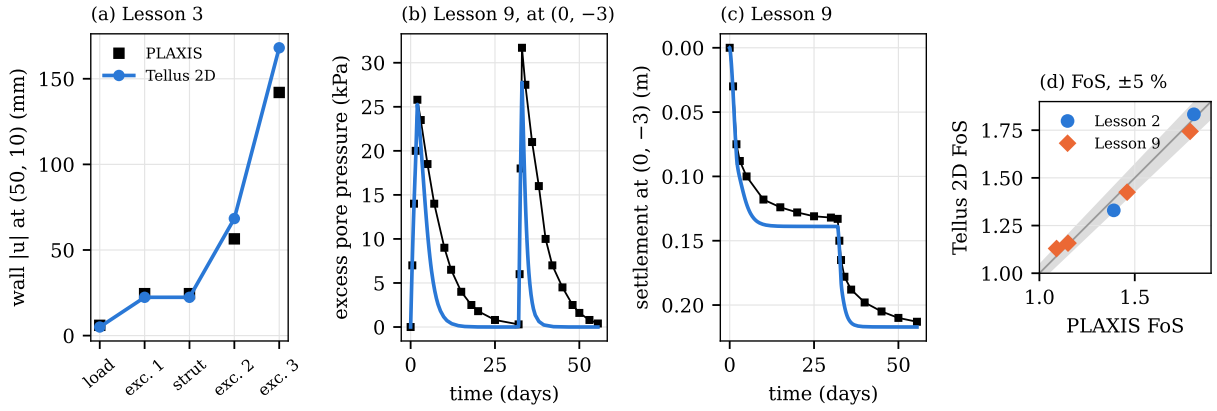


Figure 2: PLAXIS tutorial lessons: (a) Lesson 3, wall displacement at (50, 10) at the end of each phase; (b) Lesson 9, excess pore pressure under the embankment centre at (0, −3) and (c) settlement of that point against time, phases 1–4; (d) all factors of safety of Lessons 2 and 9 against PLAXIS, with a ±5% band. Black squares are PLAXIS, read off the manual's figures.

7 Field benchmarks

Table 4 summarises the three case histories, next to the Tellus3D results for the same data and the measurements, and Figure 4 shows the deformed meshes written by the program during each run. The Tellus3D study also includes a pair of EPB tunnels, which needs the staged excavation of a three-dimensional shield and is not repeated here.

7.1 Deep excavation: Sukhumvit station

The Sukhumvit station of the Bangkok MRT is a 21 m deep top-down excavation, 23 m wide, inside a 1 m diaphragm wall 27.9 m deep [19]. The half model, 71.5 m wide and 45 m deep, uses Hardening Soil for the seven layers with the paper's Table 7, undrained (type A) clays, a wall with undrained interfaces of $R_{\text{inter}} = 0.7$, the roof, access and concourse slabs and the 1.8 m base slab cast after their stages, and the centreline columns and bored piles smeared per metre with half their stiffness on the symmetry line. The pore pressures follow the deep-well drawdown profile of the paper's Figure 3, and the level inside the wall is lowered with each stage to the excavation level, which caps the profile there.

After the base slab the wall deflects 40.7 mm towards the pit at −17.9 m, against 36 mm at 19–20 m measured by inclinometer IN4, 34.6 mm at −13.2 m in the Tellus3D quarter model and 36.0 mm at −17.5 m in a plane-strain slice of that model (Figure 5a). The plane-strain results of the two programs agree within 13% in the maximum and 0.4 m

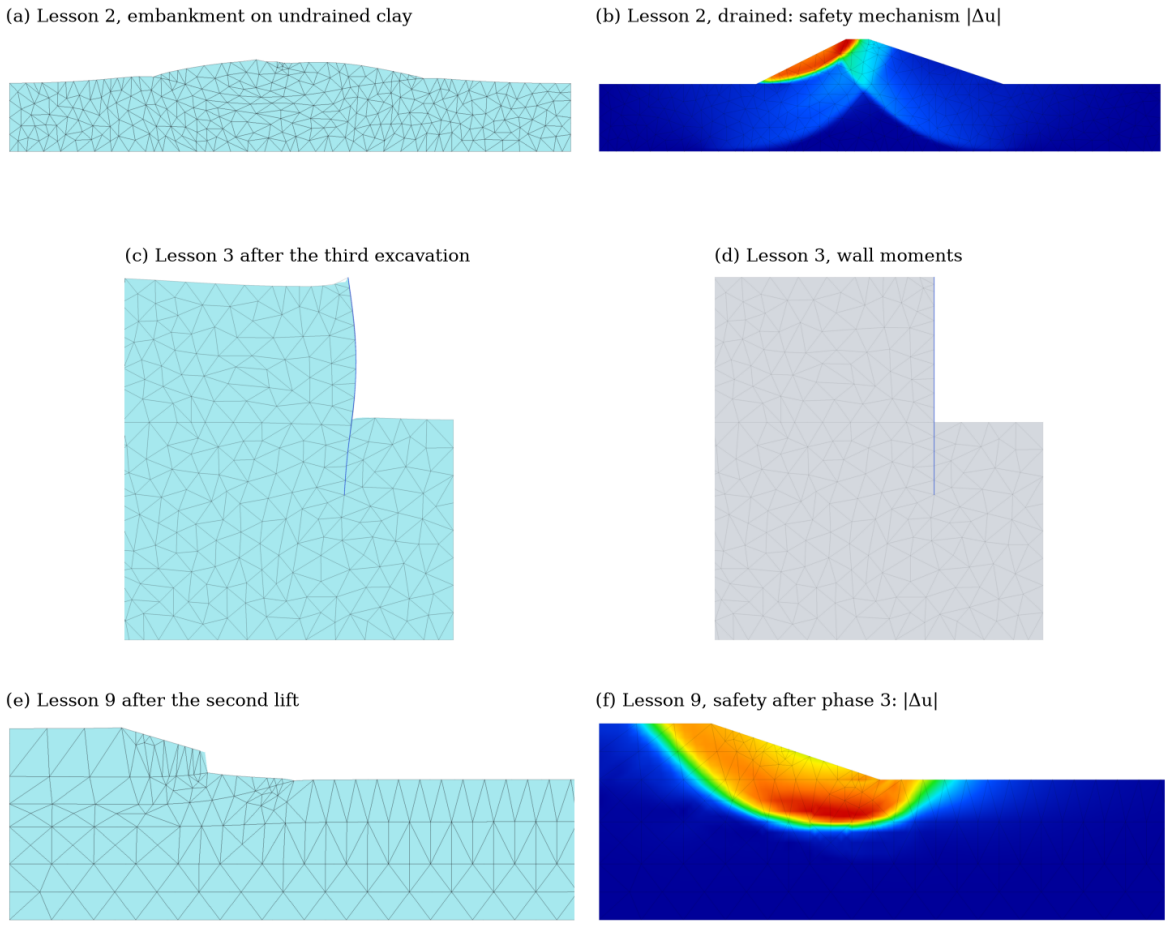


Figure 3: Pictures written by `show_picture` during the tutorial runs: (a) Lesson 2, deformed mesh after construction on undrained clay, with the heave beside the toes; (b) Lesson 2, drained, incremental displacements at failure of the safety analysis, a slip through the slope; (c) Lesson 3, deformed mesh after the third excavation, with the wall, interfaces and strut; (d) Lesson 3, bending moments in the wall; (e) Lesson 9, deformed mesh after the second lift; (f) Lesson 9, incremental displacements at failure of the safety analysis after the second lift.

Table 4: Field benchmarks in plane strain, against Tellus3D with the same data [8] and the measurements. Wall times of single runs on the cloud box, with the factorisation on the RTX 4090 (Table 6).

Case	Elements	Time	Tellus 2D	Tellus3D	Measured
Sukhumvit excavation, drawdown	4,392	51 s	wall 40.7 mm at −17.9 m	34.6 mm (3D), 36.0 mm (slice)	IN4 36 mm at 19–20 m
hydrostatic water	4,392	57 s	wall 87.8 mm at −23.0 m	–	–
Muar embankment, Table 1 set	2,889	129 s	fails at 4.07 m	fails at 4.4 m	fails at 5.5 m
Table 3 set	2,889	103 s	fails at 4.33 m	fails at 4.5 m	fails at 5.5 m
SBIA TV2, measured vacuum	3,131	170 s	0.979 m at 145 d	1.04 m	0.97 m at 144 d
calibrated vacuum	3,131	163 s	1.38 m at 145 d	1.47 m	0.97 m at 144 d

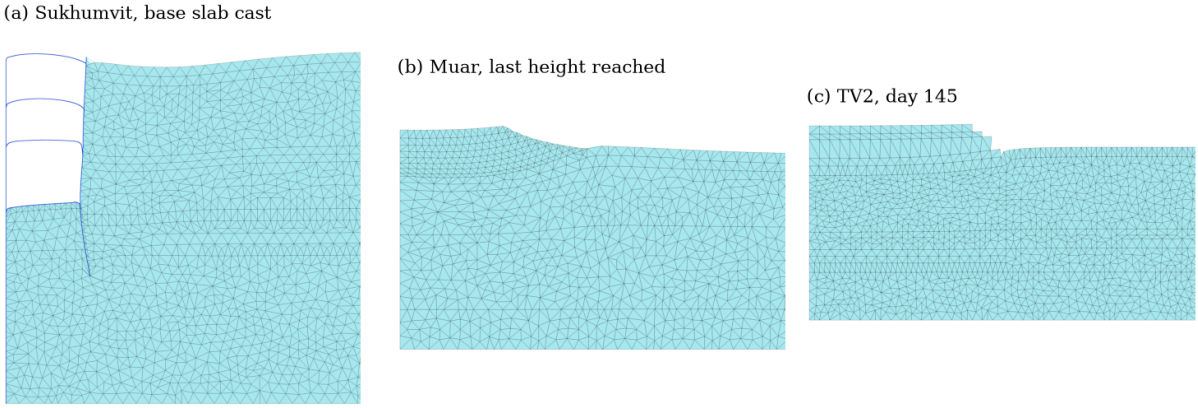


Figure 4: Deformed meshes written by `show_picture` during the field runs: (a) Sukhumvit excavation after the base slab, scaled 50 times (largest displacement 50 mm), with the wall, slabs and centreline column in blue (slabs cast during the excavation start from zero displacement while the wall they join had already moved, so they are drawn bent where they meet it); (b) Muar embankment at the last fill height reached with the Table 1 set, scaled 10 times; (c) TV2 at day 145, scaled 2 times (largest displacement 0.98 m).

in its depth, while the 3D model deflects less and higher because the corners of the box stiffen its long side. Stage 1 deflects 28.5 mm at the top against 14–18 mm measured and 25.7–27.5 mm in Tellus3D, because the temporary prop of the first stage is not modelled, and the settlement trough is narrower and deeper than array SS1, with 41.6 mm at 11 m from the wall against 26 mm and 7.3 mm at 31 m against 15 mm (Figure 5b), as in Tellus3D. With hydrostatic water the wall deflects 87.8 mm, 2.2 times as much, which agrees with the statement of Likitlersuang et al. [19] that a hydrostatic profile doubles the deflection.

This case failed three times before the changes of Section 4, and the failures show why field cases are needed. With interfaces that stayed active on the excavated side, the second stage collapsed at $\Sigma M_{\text{stage}} = 0.879$, and with that fixed it still collapsed at 0.927, because the soil beside the wall at the 7.5 m excavation level carried the full 55 kPa of the regional profile at the exposed base and went into tension at the corner of the wall. The pit is dewatered to its excavation level in the field, so capping the profile by the pit's water level is the correct input rather than a numerical device. Tellus3D passed the same stage with 55 kPa at the base, probably because its tetrahedra of about 3 m smear the stress concentration that the refined 15-node triangles along the wall resolve.

7.2 Embankment to failure: Muar

The Muar trial embankment was built on soft marine clay at 0.4 m per week until it failed at about 5.5 m of fill [20]. The half model uses Soft Soil for the four clay layers with the paper's Tables 1, 3 and 4, Mohr–Coulomb for the fill, and raises the fill in 0.5 m lifts, each placed undrained and consolidated for 8.75 days. The fill fails at 4.07 m with the Table 1 set and 4.33 m with the Table 3 set, against 4.4 and 4.5 m in Tellus3D and 5.5 m measured (Figure 6). At 4.0 m,

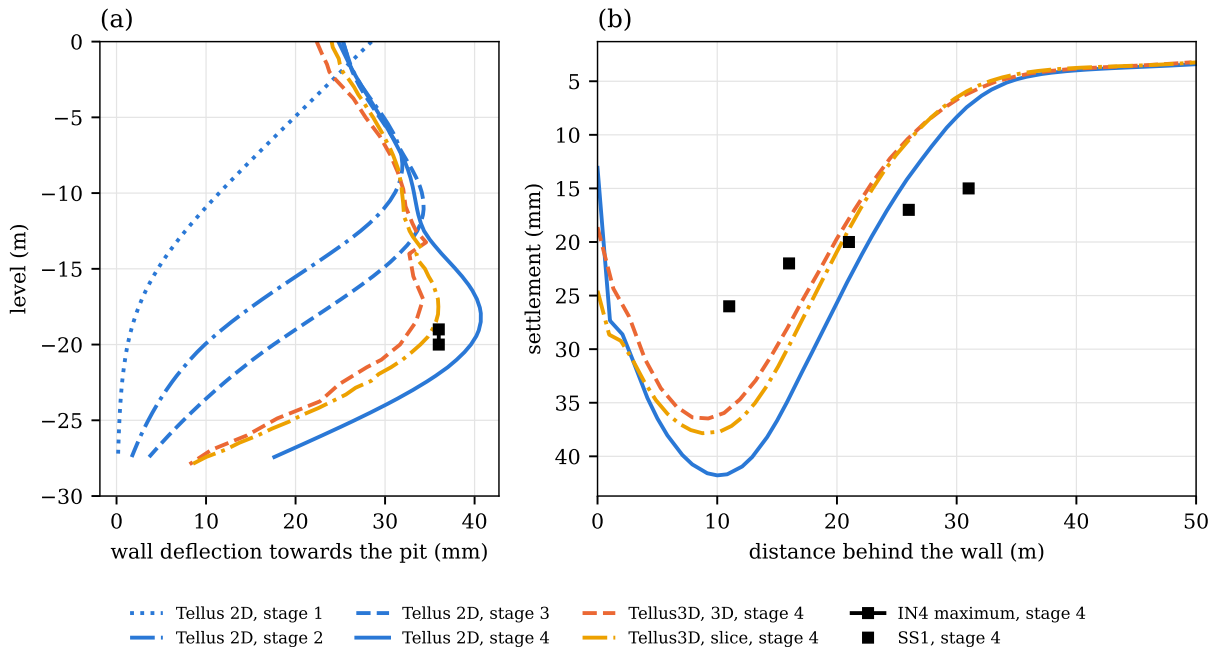


Figure 5: Sukhumvit station excavation: (a) wall deflection after each stage in Tellus 2D and after stage 4 in Tellus3D, against the measured maximum of IN4, and (b) surface settlement behind the wall after stage 4 against settlement array SS1.

the last height that both programs reach, the centre settlement of Tellus 2D is 9–10% below that of Tellus3D and the lateral displacement of inclinometer I3 within 3%, and both programs follow the measured settlement to about 2.5 m of fill, then fall behind, and over-predict the lateral displacement at low fills. The plane-strain section fails earlier than the 3D quarter model, whose embankment is 90 m long with resisting ends, and both fail about 1 m early, most likely because the Soft Soil model fails on the Mohr–Coulomb hexagon, which is weaker in plane strain than the circular section of the Cam-clay model of the original analysis. Runs of the same model during the work gave 3.95–4.09 m as the Soft Soil stress update, its iteration stiffness and the relaxation of its consolidation phases were changed, so a failure height near this limit moves by about 0.15 m with the details of the stress integration.

7.3 Vacuum preloading with vertical drains: Suvarnabhumi TV2

Test embankment TV2 combined vacuum under a membrane with PVDs and a reduced sand surcharge on soft Bangkok clay [21, 22], and its centreline settled 0.97 m in 144 days, with 0.70 m at 3 m depth and 0.35 m at 6 m. The half model uses Soft Soil for five clay layers from the Modified Cam-clay parameters of Bergado et al. [21], the site's under-hydrostatic pore pressure profile, the equivalent vertical permeability of Chai et al. [17] with Hansbo's smear and well resistance [18] in the treated zone, which raises k_v by a factor of 14 in the crust and 32–36 below, and the measured membrane vacuum, which peaks at 62 kPa near day 30. The vacuum is held in the whole sand blanket over the treated area, which carries it to the drains, because the membrane is sealed at its edge. A first model that held it only on top of the blanket left the blanket's side face draining, and the inward seepage through the permeable sand pulled that face apart.

With the measured vacuum the settlements at day 145 are 0.979, 0.691 and 0.361 m at the surface, 3 m and 6 m, against 0.97, 0.70 and 0.35 m, within 3% at every depth (Figure 7). The early settlement is too fast, as in Tellus3D but less so, with 0.344 m at day 25 and 0.698 m at day 45 against about 0.30 and 0.52 m measured and 0.39 and 0.76 m in Tellus3D, because the full membrane vacuum enters the top of the treated zone and spreads through the raised permeability, whereas the measured vacuum decays along the drains [23]. The suction at 3 m peaks at 41 kPa at day 38 against 28–32 kPa measured and 39 kPa in Tellus3D, and the edge of the treated area moves outwards by 63 mm against about 55 mm measured. The calibrated vacuum history of Rujikiatkamjorn [22], which raises the vacuum to 40 kPa by day 10, gives 1.38, 1.01 and 0.55 m at day 145, 42% more than measured at the surface, as Tellus3D over-predicts

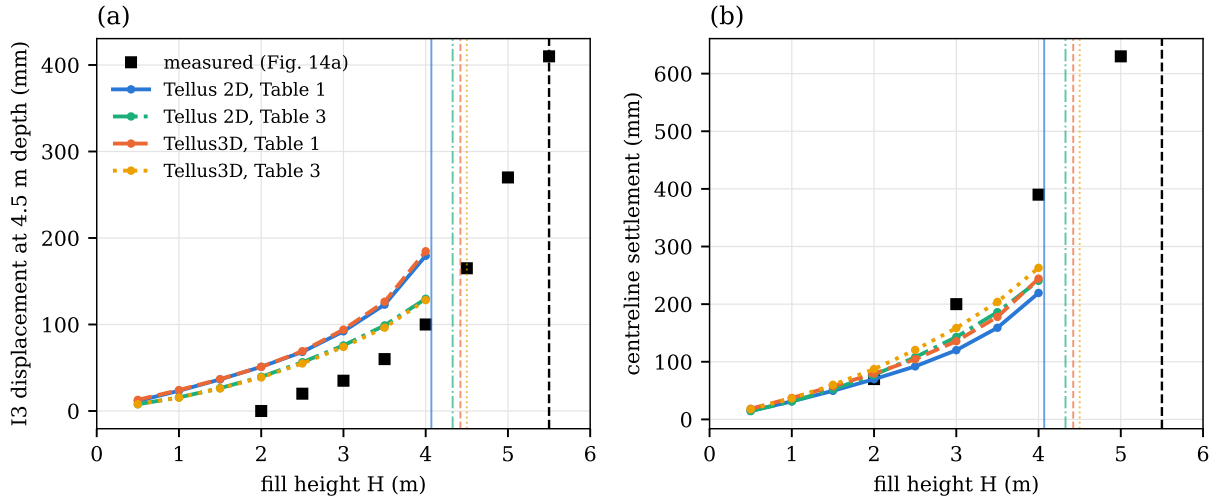


Figure 6: Muar embankment raised to failure: (a) lateral displacement of inclinometer I3 at 4.5 m depth and (b) centreline settlement against fill height, for Tellus 2D and Tellus3D with the two parameter sets. Vertical lines mark the failure heights of the models and, dashed in black, the measured one.

Table 5: The soil-cement column case in Tellus 2D against the measurements. Times of single runs on the cloud box (Table 6).

Case	Elements	Time	Tellus 2D	Measured
Lian-Yun-Gang, E_{col} = 80 MPa	5,769–9,493	99–1,107 s	0.04–0.18 m	0.03–0.54 m
E_{col} = 20 MPa	5,769–9,493	105–406 s	0.11–0.34 m	0.03–0.54 m
no columns	2,611–2,849	88–195 s	0.65–1.05 m	–

by 50% with it, so both programs show that this history, calibrated for a different plane-strain model, does not transfer. Before the relaxation of consolidation phases (Section 4) this analysis stopped at day 9.2 with the crust under the edge of the sand blanket yielding, which was first read as a physical failure.

8 A further case history: embankments on soil-cement columns

A further case tests a feature that the first three do not use, soft clay improved by soil-cement columns. It is taken from an open publication, with the parameters of the source and no tuning to the field data, and Table 5 summarises it. Figure 9 shows a deformed mesh.

The Lian-Yun-Gang section of the Xu–Lian expressway in China crosses 2 m of crust over about 8 m of soft clay improved by dry-mixed soil-cement columns of 0.5 m diameter on a triangular grid of 1.1–1.6 m spacing, which reach 0.5 m into the stiff layer under a 0.3 m lime-soil slab, and the final settlements of embankments 2–5 m high were measured at many sections [24]. The half model uses Soft Soil for the clays with $\lambda^* = C_c/(2.3(1+e))$ from the source's Figure 11, and turns the triangular grid into plane-strain walls along the embankment axis at the row spacing $S\sqrt{3}/2$, with a thickness that keeps the area ratio, Mohr–Coulomb walls with $E = 100\ q_u$ = 80 MPa and $c = q_u/2$, and a linear elastic slab. The fill rises in 0.5 m lifts, each placed undrained and consolidated for 10 days, and a final consolidation of 20,000 days gives the final settlement. A drained analysis of the normally consolidated clay stalled in the iterations, and a Mohr–Coulomb slab yielded in shear at a few points above the column edges, under up to 440 kPa, which held the error at four times the tolerance in some lifts. The slab strength does not enter the settlement, which the source calculates for a flexible slab.

With the column modulus of 100 q_u the model is two to four times too stiff (Figure 8). The final settlements are 36–64 mm at S = 1.1 m, 50–89 mm at 1.35 m and 65–176 mm at 1.6 m for fills of 2.5–4.5 m, against 30–160 mm, 60–320 mm and 190–540 mm measured. The source avoids this stiffness in its own calculation, because it takes a

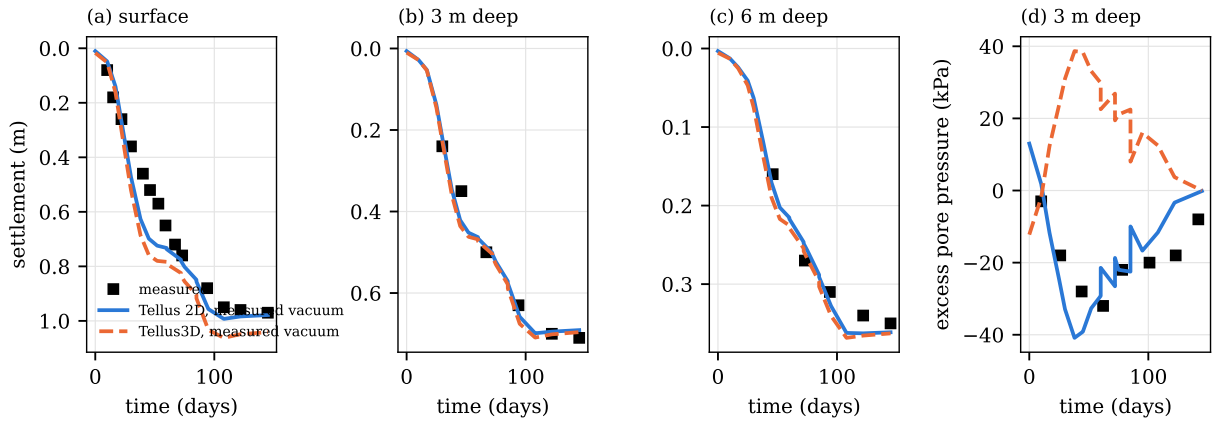


Figure 7: Vacuum-PVD preloading of TV2 under the centre of the treated area: settlement of (a) the surface, (b) a point 3 m deep and (c) a point 6 m deep, and (d) excess pore pressure 3 m deep, suction negative as in the source, for Tellus 2D and Tellus3D with the measured vacuum. Black squares are field measurements read off the published figures.

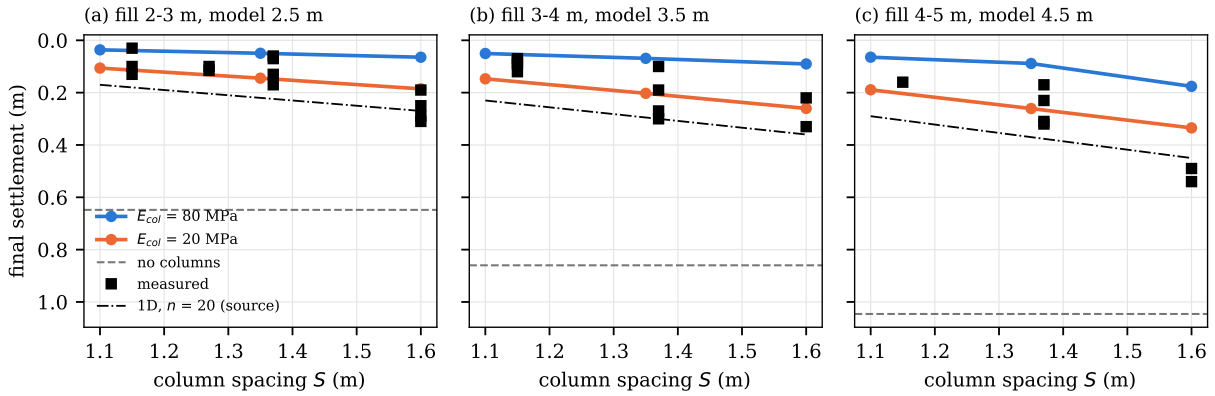


Figure 8: Lian-Yun-Gang soil-cement columns: final settlement against column spacing for fills of (a) 2–3 m, (b) 3–4 m and (c) 4–5 m, for Tellus 2D with column moduli of 80 and 20 MPa and without columns, against the field data and the one-dimensional calculation with a stress concentration ratio of 20 of Chai et al. [24].

column-to-soil modulus ratio of about 45 from the ratio of the unconfined strengths of the columns and the soil [24]. A column modulus of 20 MPa, which is 45 times the oedometer modulus of the soft clay at a vertical stress of about 50 kPa, gives 0.11–0.19, 0.15–0.26 and 0.19–0.34 m, which lies within the measured ranges for S = 1.15–1.37 m, slightly above them for 1.15 m under 3–4 m of fill, and below the two measurements of 0.49 and 0.54 m for S = 1.6 m under 4–5 m of fill. The scatter of the field data, up to a factor of three at one spacing and height, comes mainly from the thickness of the soft layer, which varies from 5.6 to 13.3 m along the section, whereas the model has 8 m. Without columns the 2.5, 3.5 and 4.5 m fills settle 0.65, 0.86 and 1.05 m, 3–18 times as much as with columns. The case therefore confirms that the plane-strain wall model of a column grid carries the load as the field does, but its prediction depends on the column modulus, which the rule $E = 100\, q_u$ overestimates for these columns by a factor of about four.

9 Computing times

The first runs of the field cases on the laptop were slow, and the reason was the factorisation. The scalar LDL^{T} took 11–15 s on one core for the 72,474 coupled equations of TV2, and a consolidation phase refactorises whenever its time step changes by more than 30% and after each failed step, so the TV2 analysis ran for 2.2 hours before it failed in its fourth phase. The supernodal factorisation ported from Tellus3D takes 2.7–4.0 s for the same matrix on the laptop's

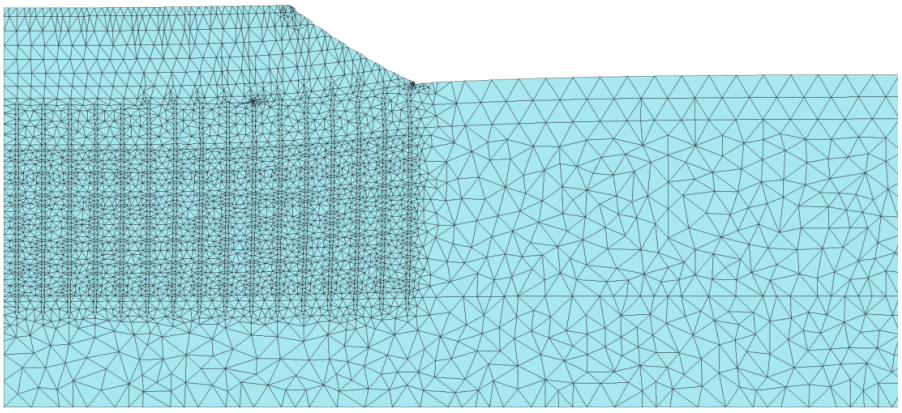


Figure 9: Deformed mesh of Lian-Yun-Gang with S = 1.35 m under 3.5 m of fill at the end of consolidation, with the column walls, scaled 20 times.

Table 6: Computing times of all runs, one after another on the cloud box (RTX 4090 for the factorisation with cuDSS, 16 threads of an AMD EPYC 7502 for the stress updates). Equations: the largest system of the run. Factorisations: the number (all of them on the GPU) and their total time. Wall: the run from the start of the MCP server to the last picture written.

Run	Elements	Equations	Factorisations	Factorisation (s)	Wall (s)
PLAXIS Lesson 2, drained + undrained	591	9,504	4	0.6	163
PLAXIS Lesson 3	706	11,613	5	0.8	11
PLAXIS Lesson 9	343	8,206	34	4.4	144
Sukhumvit, drawdown	4,392	71,025	9	7.2	51
Sukhumvit, hydrostatic	4,392	71,025	9	7.2	57
Muar to failure, Table 1	2,889	65,630	25	20.3	129
Muar to failure, Table 3	2,889	65,630	25	20.2	103
TV2, measured vacuum	3,131	74,862	73	83.9	170
TV2, calibrated vacuum	3,131	74,862	53	60.6	163
Lian-Yun-Gang, E_{col} = 80 MPa (12 runs)	2,611–9,493	227,813	571	1395.0	88–1107
Lian-Yun-Gang, E_{col} = 20 MPa (9 runs)	5,769–9,493	227,813	399	1114.3	105–406

eight cores, and cuDSS on the RTX 4090 takes 1.1–1.2 s, of which most is the analysis phase that is repeated at every refactorisation. One lesson of the first cloud runs is that a GPU should serve one process at a time, because eight processes that shared the card did not finish one factorisation in several minutes, so every timing below comes from runs made one after another.

All 30 runs of the paper were repeated in this way on a rented box with an RTX 4090 for the factorisation and 16 threads of an AMD EPYC 7502 for the stress updates, and Table 6 gives the wall time of each, the largest system of equations and the number and total time of its factorisations, all of which ran on the GPU. The 30 runs took 6,301 s, or 1.8 hours, of which the factorisation was 43%. The share of the factorisation grows with the size of the system, from 0.3–7% in the tutorial lessons, whose systems have 8,000–12,000 equations, to 13–20% in the excavation and the Muar embankment with about 70,000, 37–49% in TV2, whose consolidation phases refactorise 53–73 times, and 32–71% in the soil-cement column models, where the thin column walls refine the mesh to 138,000–228,000 equations and each factorisation takes 2.0–3.4 s. The rest of the time is the stress update of the Soft Soil and Hardening Soil models, which runs on the CPU, so a model of up to 9,500 fifteen-node triangles is calculated in at most 19 minutes and the factorisation is no longer the single bottleneck.

10 Discussion

The accuracy of Tellus 2D falls in three tiers that follow the kind of reference rather than the application (Figure 1a). The verification suite shows that each part of the program solves its equations, since all 14 results of the figure lie within 4% of the closed-form and reference solutions, and the tutorial lessons show that the parts work together as in an established program on fully specified input, since 9 of their 13 results lie within 10% of PLAXIS and the factors of safety within 4.3%. The four tutorial results outside 10% are explained by features that the two programs treat differently, interfaces against undrained Hardening Soil clay in Lesson 3 and the permeability of soft soil above the water table in Lesson 9. The field cases form the third tier, with ratios from 0.25 to 1.6, and they show how far published parameters and the program's models represent real ground rather than how well the code solves its equations. A stability or consolidation problem is therefore not harder for the program than a foundation problem, whereas a field case is harder than a benchmark of either kind, because its inputs are uncertain.

Two observations place the field errors in the inputs and the soil models rather than in the code. Wherever Tellus 2D and Tellus3D model the same case with the same data they agree closely, the excavation within 13% and the embankment within 10% with the plane-strain fill failing 0.2–0.3 m earlier, and both make the same errors against the field, including the early failure of the Muar embankment, which most likely comes from the Mohr–Coulomb failure surface of the Soft Soil model, weaker in plane strain than the Cam-clay circle of the original analysis. In the soil-cement column case, the outcome depended on one input that the published data do not fix, the field modulus of the columns, and with the source's own column-to-soil modulus ratio the settlements fall within the field scatter, which itself spans a factor of three because the soft layer varies from 5.6 to 13.3 m in thickness. This is the usual position of a back-analysis of an improved embankment, and it shows that the program reproduces the mechanics once the uncertain inputs are fixed, not that those inputs can be predicted from the published data.

The column case also shows the limit of the numerics. It ran only after a modelling choice that a reader should know, a linear elastic lime-soil slab, which removed a few stress points that held the iterations at about four times the tolerance without a failure mechanism, so robust convergence at such points beside a stiffness contrast, by a consistent tangent or a local substepping with error control, is the main numerical work that remains. Near-failure results are sensitive in the same way, since the failure height of the Muar embankment moved by about 0.15 m between versions of the stress integration and the iteration, which is about 4% of the height and should be read as the resolution of such a result.

The most useful outcome of the calibration is that it found four defects that 75 verification tests had missed, each of which would have reached a design analysis unnoticed. The interface defect affected every excavation with interfaces, the default for a plate in Tellus 2D, yet a footing next to a wall never excavates the soil on one side of the plate. The mesher defect needed a line load along part of an inclined cluster edge, and the safety defect needed a phase before it that left its out-of-balance at a few failing nodes or as an unloading, which consolidation phases and Hardening Soil footings do, whereas the verification tests began their safety analyses from well-balanced states. The relaxation defect needed long consolidation phases with a few stress points at a stiffness contrast, which the verification tests and the first three case histories did not have. These are combinations of features, which simple tests miss and staged problems exercise, so comparisons on fully specified staged problems, such as the PLAXIS tutorial lessons, should be part of the verification of a geotechnical code. The interface and mesher defects are now covered by regression tests, and the safety and relaxation defects by the tutorial and case scripts. Running the suite on the GPU box also exposed a test that required a recalculated phase to repeat its result bit for bit, which the GPU factorisation does not do because it sums in no fixed order, so that test now compares to round-off. The corrected start of a safety analysis changed one result without a reference: the dry sand slope of the chat assistant's tests (3 m at 1:2, $\varphi' = 32°$, $\psi = 2°$, after gravity loading) needs 300 instead of 100 strength steps to reach failure and gives 1.47 instead of 1.38, so that test fails at the default of 100 steps and remains open, because neither value can be confirmed without an independent reference for this slope.

The GPU solver changes what such a calibration costs. All 30 analyses ran one after another in 1.8 hours on one rented RTX 4090, so the whole set can be repeated after every change to the program, and a parametric study such as the 21 runs of the soil-cement columns takes about 1.5 hours. The factorisation now takes 43% of the time, and the stress updates of the Soft Soil and Hardening Soil models most of the rest, so faster stress integration on the CPU or the GPU would now pay as much as a faster factorisation. A GPU should serve one analysis at a time, because the first cloud runs, with eight analyses sharing one card, did not finish a single factorisation in several minutes.

The work was done by an LLM agent that drove the program through its MCP server, wrote the case scripts, diagnosed the failures with pictures, probe runs and dumps of the stress points at the worst node, changed the program and its

tests, rented and destroyed the cloud box and drew the figures, while the author directed it and checked the results. This mode of work is the one the program is designed for, and the calibration shows its strength and its limits. The agent found the defects because it could run many probe models cheaply, but it also proposed a fix that the regression tests rejected, a looser convergence reference for safety analyses that doubled the factor of safety of the Prandtl strip, and it misread results in both directions, first a correct physical result, the tension at the base of a pit under the regional pore pressure, as a numerical problem, and then a numerical stall, the TV2 analysis that stopped at day 9.2, as a physical failure of the crust. Four case histories and three tutorial lessons are still too few to establish the program's accuracy in general, the reference values read off figures carry an uncertainty of a few percent, and the plane-strain walls that represent a column grid are an idealisation whose stiffness against the three-dimensional grid has not been checked. The tests and the engineer's judgement on the inputs and on whether a result is physical remain the parts of the work that the agent cannot replace.

11 Conclusions

Tellus 2D, a plane-strain geotechnical FE program designed for an AI agent as its primary user, was calibrated in eight applications, from bearing capacity, slope stability, consolidation, groundwater flow and piles to deep excavation, vacuum preloading and soil-cement columns, against 82 verification tests, three lessons of the PLAXIS 2D Tutorial Manual and four instrumented case histories in soft clay, with every model built and run headless through the program's MCP server and the published parameters used without tuning. The program solves its equations correctly, because all verification tests pass and the closed-form and reference solutions are reproduced within 4%. It also combines its parts as an established program does, because the factors of safety of the tutorial lessons agree with PLAXIS within 4.3% and the wall displacements of the strutted excavation within 9–17% until it is submerged. The two tutorial differences that remain, 18–21% larger wall displacements after the submerged stages and a pore pressure dissipation about three times faster than in PLAXIS, are traced to weak interfaces against undrained Hardening Soil clay and to the permeability of soft soil above the water table, which the two programs treat differently.

Against field measurements the accuracy depends on how well the source fixes the inputs rather than on the code. With fully published data the plane-strain models reproduce the wall deflection of a 21 m Bangkok excavation within 13% and the final settlements of the Suvarnabhumi vacuum trial within 3% at three depths, and they fail the Muar embankment 1.2–1.4 m early, in every case close to Tellus3D on the same data, which places the remaining error in the soil models and their parameters. Where one input is uncertain, it governs the outcome, since the settlements of the soil-cement columns at Lian-Yun-Gang fall within the measured scatter with the column-to-soil modulus ratio of the source but are two to four times too small with the common rule $E = 100\ q_u$. The column case also needed an elastic lime-soil slab to converge, so robust convergence at a few stress points beside a stiffness contrast remains the main numerical work for the program.

The calibration was most useful as a search for defects. It found and fixed four that the existing 75 tests had missed, in the interfaces of excavated sides, the meshing of lines along cluster edges, the start of safety analyses and the relaxation of consolidation phases, and three of them had made benchmarks of this paper fail, including a vacuum analysis whose numerical stall was first read as a physical failure. It also added the features that the cases needed, measured pore pressure profiles, vacuum preloading, open water on walls and undrained interfaces, and the supernodal GPU solver of Tellus3D, with which all 30 analyses of the paper, of up to 9,500 elements and 228,000 equations, ran in 1.8 hours on one RTX 4090. Comparisons on fully specified staged problems should therefore be part of the verification of any geotechnical code, and an agent that drives the program through typed tools makes such comparisons cheap enough to repeat after every change, while the engineer's judgement of the inputs and of whether a result is physical remains the part of the work that the agent cannot replace.

Code and data availability

The source code of Tellus 2D, its tests, the case and tutorial scripts, their results, the digitised PLAXIS references and the figure script are available from the author on request (repository `https://github.com/Sompote/Tellus`).

Acknowledgements

The case scripts, the program changes and the figures were written by an LLM agent (Claude Code, Anthropic) driving the program through its MCP server, and the author checked the models, the results and the text.

References

[1] Bentley Systems. *PLAXIS 2D 2025.1 Tutorial Manual*. https://files.seequent.com/PLAXIS/Manuals/PLAXIS_2D/English/PLAXIS_2D_1_Tutorial%20Manual.pdf, 2025.

[2] NVIDIA. cuDSS: NVIDIA CUDA Direct Sparse Solver, documentation, version 0.8. https://docs.nvidia.com/cuda/cudss/, 2025.

[3] I. S. Duff and J. K. Reid. The multifrontal solution of indefinite sparse symmetric linear equations. *ACM Transactions on Mathematical Software*, 9(3):302–325, 1983. doi:10.1145/356044.356047.

[4] J. W. H. Liu. The role of elimination trees in sparse factorization. *SIAM Journal on Matrix Analysis and Applications*, 11(1):134–172, 1990. doi:10.1137/0611010.

[5] Anthropic. Model Context Protocol specification. https://modelcontextprotocol.io, 2024.

[6] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023. doi:10.48550/arXiv.2210.03629.

[7] D. M. Potts and L. Zdravković. *Finite Element Analysis in Geotechnical Engineering: Application*. Thomas Telford, London, 2001. doi:10.1680/feaigea.27831.

[8] S. Youwai. Tellus3D: GPU-accelerated three-dimensional geotechnical finite element analysis designed for AI agents. Preprint, 2026.

[9] S. W. Sloan and M. F. Randolph. Numerical prediction of collapse loads using finite element methods. *International Journal for Numerical and Analytical Methods in Geomechanics*, 6(1):47–76, 1982. doi:10.1002/nag.1610060105.

[10] T. Schanz, P. A. Vermeer, and P. G. Bonnier. The hardening soil model: formulation and verification. In *Beyond 2000 in Computational Geotechnics*, pages 281–296. Balkema, Rotterdam, 1999 (reissued by CRC Press, 2019). doi:10.1201/9781315138206-27.

[11] T. Benz. *Small-strain stiffness of soils and its numerical consequences*. PhD thesis, University of Stuttgart, 2007.

[12] P. A. Vermeer and H. P. Neher. A soft soil model that accounts for creep. In *Beyond 2000 in Computational Geotechnics*, pages 249–261. Balkema, Rotterdam, 1999 (reissued by CRC Press, 2019). doi:10.1201/9781315138206-24.

[13] P.-A. von Wolffersdorff. A hypoplastic relation for granular materials with a predefined limit state surface. *Mechanics of Cohesive-frictional Materials*, 1(3):251–271, 1996. doi:10.1002/(SICI)1099-1484(199607)1:3<251::AID-CFM13>3.0.CO;2-3.

[14] M. A. Biot. General theory of three-dimensional consolidation. *Journal of Applied Physics*, 12(2):155–164, 1941. doi:10.1063/1.1712886.

[15] K. Terzaghi. *Theoretical Soil Mechanics*. John Wiley & Sons, New York, 1943. doi:10.1002/9780470172766.

[16] D. V. Griffiths and P. A. Lane. Slope stability analysis by finite elements. *Géotechnique*, 49(3):387–403, 1999. doi:10.1680/geot.1999.49.3.387.

[17] J.-C. Chai, S.-L. Shen, N. Miura, and D. T. Bergado. Simple method of modeling PVD-improved subsoil. *Journal of Geotechnical and Geoenvironmental Engineering*, 127(11):965–972, 2001. doi:10.1061/(ASCE)1090-0241(2001)127:11(965).

[18] S. Hansbo. Consolidation of fine-grained soils by prefabricated drains. In *Proceedings of the 10th International Conference on Soil Mechanics and Foundation Engineering*, volume 3, pages 677–682, Stockholm, 1981.

[19] S. Likitlersuang, C. Surarak, D. Wanatowski, E. Oh, and A. Balasubramaniam. Finite element analysis of a deep excavation: a case study from the Bangkok MRT. *Soils and Foundations*, 53(5):756–773, 2013. doi:10.1016/j.sandf.2013.08.013.

[20] B. Indraratna, A. S. Balasubramaniam, and S. Balachandran. Performance of test embankment constructed to failure on soft marine clay. *Journal of Geotechnical Engineering*, 118(1):12–33, 1992. doi:10.1061/(ASCE)0733-9410(1992)118:1(12).

[21] D. T. Bergado, J.-C. Chai, N. Miura, and A. S. Balasubramaniam. PVD improvement of soft Bangkok clay with combined vacuum and reduced sand embankment preloading. *Geotechnical Engineering Journal (SEAGS)*, 29(1):95–121, 1998.

[22] C. Rujikiatkamjorn. *Radial consolidation modelling incorporating vacuum preloading*. PhD thesis, University of Wollongong, 2005.

[23] B. Indraratna, C. Rujikiatkamjorn, and I. Sathananthan. Analytical and numerical solutions for a single vertical drain including the effects of vacuum preloading. *Canadian Geotechnical Journal*, 42(4):994–1014, 2005. doi:10.1139/t05-029.

[24] J.-C. Chai, S.-Y. Liu, and Y.-J. Du. Field properties and settlement calculation of soil-cement column improved soft subsoil – a case study. *Lowland Technology International*, 4(2):51–58, 2002.

https://cot.unhas.ac.id/journals/index.php/ialt_lti/article/view/307.